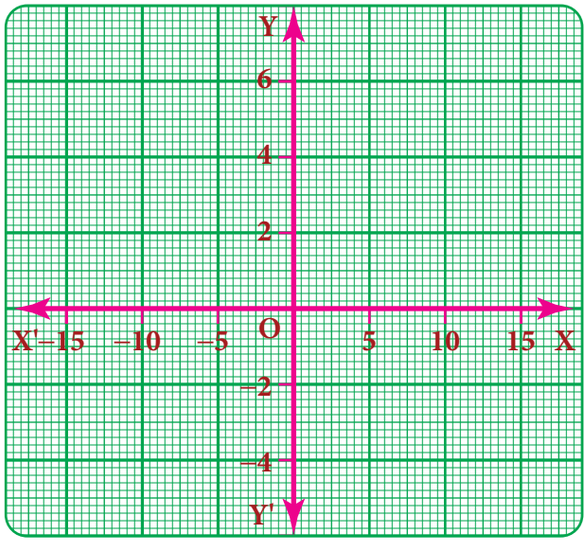
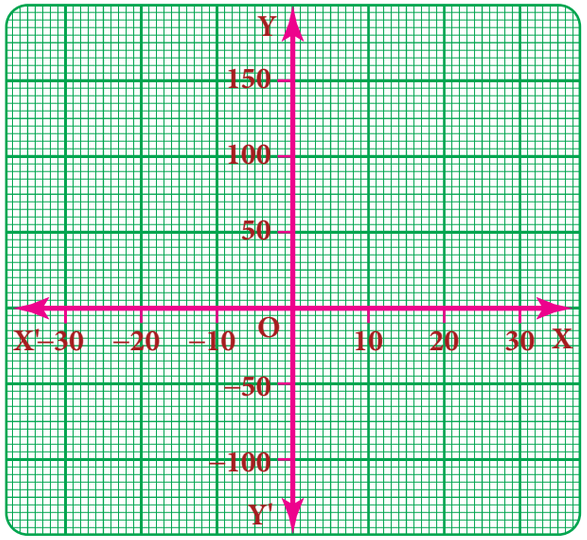
we use the concept of scale in both the axes as per the need. Represent a convenient scale at the right side corner of the graph. A few examples are given below.



Example 1
On the X axis 1 cm = 5 units.
On the Y axis 1 cm = 2 units.

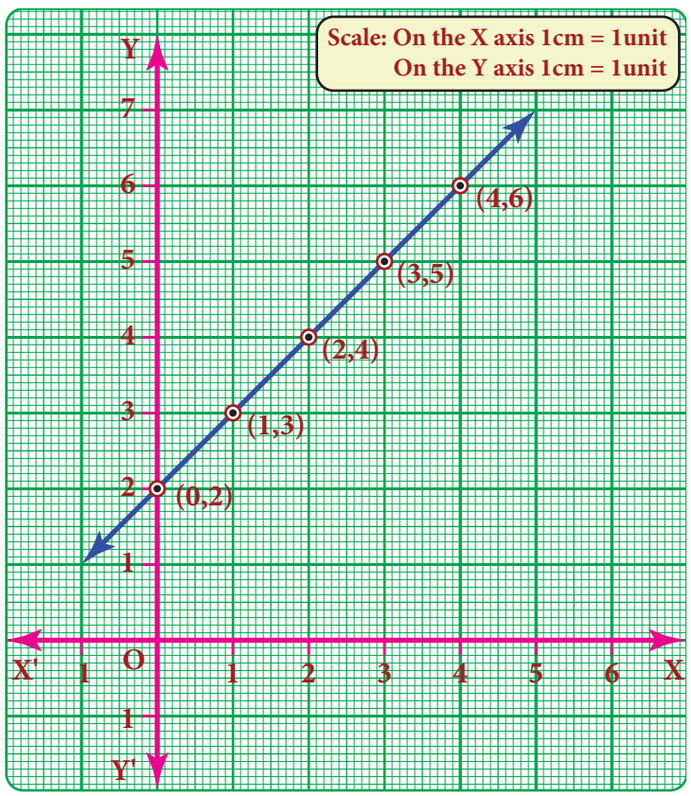


Example 2
On the X axis 1 cm = 10 units.
On the Y axis 1 cm = 50 units.

3.10 Linear Graph

3.10.1 Linear pattern

Plot the following points on a coordinate plane: (0,2), (1,3), (2,4), (3,5), (4, 6). What do you find? They all lie on a line! There is some pattern in them. Look at the y- coordinate in each ordered pair: 2 = 0+2; 3 = 1+2; 4 = 2+2; 5 = 3+2; 6 = 4+2 In each pair, the y- coordinate is 2 more than the x- coordinate. The coordinates of each point have the same relationship between them. All the points plotted lie on a line!



In such a case, when all the points plotted lie on a line, we say '*a linear pattern*' exists.

In this example, we found that in each ordered pair y value = x value +2. Therefore the linear pattern above can be denoted by the algebraic equation $y = x + 2$. Such an equation is called a *linear equation* and the line graph for linear equation is called a *linear graph*.

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